

Zulatomol solution for injection or infusion 0.5mg/ml (Salbutamol solution for injection or infusion 0.5mg/ml)

COMPOSITION:

Each ml contains Salbutamol Sulphate BP equivalent to Salbutamol 0.5mg/ml

Description: Salbutamol Injection BP is a colourless solution containing Salbutamol Sulphate BP equivalent to 0.5 mg of Salbutamol per Ampoule. The product is packed as 1mL (0.5mg) fill in 1mL clear glass ampoule.

Description after dilution: Salbutamol Injection 0.5 mg/ml on dilution with diluents 0.9% sodium chloride injection and 5% dextrose injection OR water for injection is clear colorless solution stored at 30°C and shelf life is up to 24 hours.

PHARMACEUTICAL FORM:

Solution for injection or infusion

CLINICAL PARTICULARS

INDICATIONS

Salbutamol is a selective β_2 adrenoceptor agonist. At therapeutic doses it acts on the β_2 adrenoceptors in the bronchi and uterus, with little or no action on the β_1 adrenoceptors of the heart. It is suitable for the management of an asthmatic attack, and for uncomplicated premature labour, under the direction of a physician.

Bronchodilators should not be the only or main treatment in patients with severe or unstable asthma. Severe asthma requires regular medical assessment as death may occur. Patients with severe asthma have constant symptoms and frequent exacerbations, with limited physical capacity, and PEF values below 60% predicted at baseline with greater than 30% variability, usually not returning entirely to normal after a bronchodilator. These patients will require high dose inhaled (e.g. >1mg/day beclomethasone dipropionate) or oral corticosteroid therapy.

Sudden worsening of symptoms may require increased corticosteroid dosage which should be administered under urgent medical supervision.

Relief of severe bronchospasm associated with asthma or bronchitis and for the treatment of status asthmaticus.

Management of uncomplicated premature labour in the last trimester of pregnancy.

POSOLOGY AND METHOD OF ADMINISTRATION

Method of administration

Salbutamol Injection may be administered by the subcutaneous (SC), intramuscular (I.M.) and intravenous (I.V.) injection and infusion.

Salbutamol has a duration of action of 4 to 6 hours in most patients.

Salbutamol parenteral preparations are to be used under the direction of a physician. Increasing use of beta-2 agonists may be a sign of worsening asthma. Under these conditions a reassessment of the patient's therapy plan may be required and concomitant glucocorticosteroid therapy should be considered.

Note: The contents of the ampoules of Salbutamol Solution for Intravenous Infusion must not be injected undiluted. The concentration should be reduced by 50% before administration. Salbutamol parenteral preparations should not be administered in the same syringe or infusion as any other medication.

Adults

In severe bronchospasm and status asthmaticus

Subcutaneous Route: 500 micrograms (8 micrograms /kg bodyweight) and repeated every four hours as required.

Intramuscular Route: 500 micrograms (8 micrograms /kg bodyweight) and repeated every four hours as required.

Intravenous Route: 250 micrograms (4 micrograms /kg bodyweight) injected slowly. If necessary, the dose may be repeated.

Infusion:

In status asthmaticus, infusion rates of 3 to 20 micrograms per minute are generally adequate but in patients with respiratory failure, higher dosage has been used with success. A starting dose of 5 micrograms per minute is recommended with appropriate adjustment in dosage according to patient response.

A suitable solution for infusion may be prepared by diluting 10mL (5mg) of Salbutamol Solution for Intravenous Infusion in 500mL of an infusion solution such as 0.9% sodium chloride injection and 5% dextrose injection BP to provide a salbutamol dose of 10 µg (0.01mg)/mL of solution.

In the management of premature labour

For this indication Salbutamol Solution for Intravenous Infusion is recommended using a solution prepared as above. Infusion rates of 10-45 micrograms per minute are generally adequate to control uterine contractions but greater or lesser infusion rates may be required according to the strength and frequency of contractions. A starting rate of 10 micrograms per minute is recommended, increasing the rate at 10-minute intervals until there is evidence of patient response shown by diminution in strength, frequency or duration of contractions.

Thereafter the infusion rate may be increased slowly until contractions cease. Careful attention should be given to cardio-respiratory function and fluid balance monitoring. The maternal pulse rate should be monitored and the infusion rate adjusted to avoid excessive maternal heart rates (above 140 beats per minute). Treatment discontinuation should be considered should signs of pulmonary oedema or myocardial ischaemia develop.

Once uterine contractions have ceased the infusion rate should be maintained at the same level for one hour and then reduced by 50% decrements at 6-hourly intervals. Treatment may be continued orally with Salbutamol Tablets 4 milligram given three or four times daily. As an alternative procedure or to counteract inadvertent over dosage with oxytocic drugs, Salbutamol Injection may be administered as a single injection by the intravenous or intramuscular routes. The usual recommended dose is 100 to 250 micrograms of salbutamol. The dose may be repeated according to the response of the patient.

Children

At present there is insufficient evidence to recommend a dosage regimen for routine use in children.

CONTRAINDICATIONS

Salbutamol parenteral preparations are contraindicated in patients with a history of hypersensitivity to any of their components.

Although intravenous Salbutamol used in the management of premature labour, uncomplicated by conditions such as placenta praevia, ante-partum haemorrhage or toxemia of pregnancy, Salbutamol presentations should not be used for threatened abortion.

SPECIAL WARNINGS AND SPECIAL PRECAUTIONS FOR USE

The management of asthma should normally follow a stepwise programme, and patient response should be monitored clinically and by lung function tests.

Increasing use of short acting inhaled β agonists to control symptoms indicates deterioration of asthma control. Under 2 these conditions, the patient's therapy plan should be reassessed. Sudden and progressive deterioration in asthma control is potentially life threatening and consideration should be given to starting or increasing corticosteroid therapy. In patients considered at risk, daily peak flow monitoring may be instituted.

The use of Salbutamol parenteral preparations in the treatment of severe bronchospasm or status asthmaticus does not obviate the requirement for glucocorticoid steroid therapy as appropriate.

When practicable, administration of oxygen concurrently with parenteral Salbutamol is recommended, particularly when it is given by intravenous infusion to hypoxic patients. In common with other β -adrenoceptor agonists, Salbutamol can induce reversible metabolic changes such as reversible hypokalaemia and increased blood glucose levels. The diabetic patient may be unable to compensate for this and the development of ketoacidosis has been reported. Concurrent administration of corticosteroids can exaggerate this effect.

Potentially serious hypokalaemia may result from β agonist therapy mainly from parenteral and nebulized administration. 2 Particular caution is advised in acute severe asthma as this effect may be potentiated by concomitant treatment with xanthine derivatives, steroids, diuretics and by hypoxia. It is recommended that serum potassium levels are monitored in such situations.

Diabetic patients and those concurrent receiving corticosteroids should be monitored frequently during intravenous infusion of salbutamol so that remedial steps (e.g an increase in insulin dosage) can be taken to counter any metabolic range occurring. For these patients salbutamol solution for intravenous infusion should be diluted with sodium chloride injection BP rather than sodium chloride and dextrose injection BP. Other alternative/ product with same strength & API can be used for this condition.

Lactic acidosis has been reported very rarely in association with high therapeutic doses of intravenous and nebulised short-acting beta-agonist therapy, mainly in patients being treated for an acute asthma exacerbation (see Adverse Reaction section). Increase in lactate levels may lead to dyspnoea and compensatory hyperventilation, which could be misinterpreted as a sign of asthma treatment failure and lead to inappropriate intensification of short-acting beta- agonist treatment. It is therefore recommended that patients are monitored for the development of elevated serum lactate and consequent metabolic acidosis in this setting. Salbutamol should be administered cautiously to patients with thyrotoxicosis.

Tocolysis: Serious adverse reactions including death have been reported of salbutamol to women in labour. In the mother, these include increased heart rate, transient hyperglycaemia, hypokalaemia, cardiac arrhythmias, pulmonary oedema and myocardial ischaemia. Increased fetal heart rate and neonatal hypoglycaemia may occur as a result of maternal administration.

As maternal pulmonary oedema and myocardial ischaemia have been reported during or following treatment of premature labour with β agonists, careful attention should be given to fluid balance and cardio-respiratory function, 2 including ECG should be monitored. If signs of pulmonary oedema or myocardial ischaemia develop, discontinuation of treatment should be considered.

Due to the risk of pulmonary oedema and myocardial ischaemia that has been observed, during the use of beta2-agonists in the treatment of premature labour, before these products are given to any patient with known heart disease, an adequate assessment of the patients' cardiovascular status should be made by a physician experienced in cardiology.

Cautious use of salbutamol injections is required in pregnant patients when it is given for relief of bronchospasm so as to avoid interference with uterine contractibility. During IV infusion of salbutamol, the maternal pulse rate should be monitored and not normally allowed to exceed a steady rate of 140 beats per minute.

DRUG INTERACTION

Salbutamol and non-selective beta-blocking drugs such as propranolol, should not usually be prescribed together.

USE DURING PREGNANCY & LACTATION

Administration of drugs during pregnancy should only be considered if the expected benefit to the mother is greater than any possible risk to the foetus.

During worldwide marketing experience, rare cases of various congenital anomalies, including cleft palate and limb defects have been reported in the offspring of patients being treated with salbutamol. Some of the mothers were taking multiple medications during their pregnancies. Because no consistent pattern of defects can be discerned, and baseline rate for congenital anomalies is 2-3%, a relationship with salbutamol use cannot be established.

As salbutamol is probably secreted in breast milk its use in nursing mothers is not recommended unless the expected benefits outweigh any potential risk. It is not known whether salbutamol in breast milk has a harmful effect on the neonate.

EFFECTS ON ABILITY TO DRIVE AND TO USE MACHINES

None known

SIDE EFFECTS / ADVERSE REACTIONS

Adverse reactions are listed below by system organ class and frequency.

Immune system disorders

Very rare: Hypersensitivity reactions including angioedema, urticaria, bronchospasm, hypotension and collapse.

Metabolism and nutrition disorders

Rare : Hypokalaemia.

Potentially serious hypokalaemia may result from beta-agonist therapy.

Unknown: Lactic acidosis

Lactic acidosis has been reported very rarely in patients receiving high therapeutic doses of intravenous and nebulized salbutamol therapy for the treatment of acute asthma exacerbation.

Nervous system disorders

Very common: Tremor.

Common: Headache.

Very rare: Hyperactivity.

Cardiac disorders

Very common: Tachycardia.

Common: Palpitations.

Rare: Cardiac arrhythmias including atrial fibrillation, supraventricular tachycardia and extrasystoles.

Unknown: Myocardial ischaemia.

Vascular disorders

Rare: Peripheral vasodilatation.

Respiratory, thoracic and mediastinal disorders

Uncommon: Pulmonary oedema.

In the management of pre-term labour, Salbutamol Injection has uncommonly been associated with pulmonary oedema.

Patients with predisposing factors including multiple pregnancy, fluid overload infection and pre-eclampsia may have an increased risk of developing pulmonary oedema.

Gastrointestinal disorders

Unknown: Nausea, vomiting.

In the management of premature labour, intravenous infusion of Salbutamol Injection has very rarely been associated with nausea and vomiting.

Musculoskeletal and connective tissue disorders

Common: Muscle cramps.

Injury, poisoning and procedural complications

Unknown: Slight pain or stinging on i.m. use of undiluted injection.

OVERDOSE:

The most common signs and symptoms of overdose with salbutamol are transient beta agonist pharmacologically mediated events (see Warnings and Precautions and adverse reactions)

Hypokalaemia may occur following overdose with Salbutamol. Serum potassium levels should be monitored. Nausea, vomiting and hyperglycaemia have been reported, predominantly in children and when salbutamol overdose has been taken via the oral route.

Treatment of over dosage:

Consideration should be given to discontinuation of treatment and appropriate symptomatic therapy such as cardioselective beta-blocking agents in patients presenting with cardiac symptoms (e.g. tachycardia, palpitations). Beta blocking drugs should be used with caution in patients with a history of bronchospasm.

PHARMACOLOGY:**Pharmacodynamic properties**

Salbutamol is a selective β_2 -adrenoceptor agonist. At therapeutic doses it acts on β_2 -adrenoceptors of bronchial muscle providing short-acting (4-6 hour) bronchodilation with a fast onset (within 5 minutes) in reversible airways obstruction.

Pharmacokinetic properties

Salbutamol administered intravenously has a half-life of 4 to 6 hours and is cleared partly renally and partly by metabolism to the inactive 4'-O-sulphate (phenolic sulphate) which is also excreted primarily in the urine. The faeces are a minor route of excretion. The majority of a dose of salbutamol given intravenously, orally or by inhalation is excreted within 72 hours. Salbutamol is bound to plasma proteins to the extent of 10%.

Preclinical safety data

Not applicable.

Incompatibilities

The excipients and container have no deleterious effect on the chemical stability of the active ingredient. This has been demonstrated by the stability studies carried out on the finished product.

Shelf life

36 months

After dilution shelf life:

The shelf life of the diluted product is up to 24 hrs stored at 30°C.

Special precautions for storage:

Store below 30°C and protect from light.

Storage condition after dilution:

Salbutamol Injection 0.5 mg/ml on dilution with diluents water for injection OR 0.9% sodium chloride injection and 5% dextrose injection stored at 30°C and shelf life is up to 24 hours.

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user.

Nature & contents of container:

USP Type I Clear Glass Ampoule. 10 ampoules packed in HIPs tray such tray packed in outer carton box along with Package Insert.

Manufactured by :

Gland Pharma Limited,
Unit II, Plot No. 42-52, Sy No. 166, 171, 172, & 177,
Phase III, TSIIC, Pashamylaram Village,
Patacheru Mandal, Sangareddy District-
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Product registration holder & importer:

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